



MUHAMMAD RAFLY

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Github : <https://github.com/muhammadradydocs/engineering-portfolio>

PROFESSIONAL SUMMARY

Analytical and detail-oriented Electronic and Embedded Systems Engineer with hands-on experience in designing, developing, testing, and troubleshooting embedded electronic systems, industrial automation, and robotics applications. Experienced in developing Proof-of-Concept (PoC) solutions using ESP32, Arduino, Jetson Nano, sensors, and communication modules. Skilled in system analysis, electronics integration, technical documentation, validation testing, and root cause analysis. Capable of collaborating effectively with multidisciplinary teams in research and product development environments.

EDUCATION

Bachelor of Engineering Physics with Cum Laude Predicate

Telkom University

August 2020 - August 2024

- **GPA** : 3.67/4.00.
- **Relevant Coursework** : Embedded Systems, Electronics, Analog and Digital Electronics, Instrumentation and Control Systems, Sensor Technology, Industrial Automation, Electrical Systems, PLC Systems, Data Acquisition, Communication Systems, System Integration and Validation, Technical Documentation, and Non-Destructive Testing (NDT).
- **Capstone Design** : Mobilization of Food Delivery Robot Based on Sensor Fusion and Object Detection.
- **Paper/Conference** : Utilization of Luminosity Sensors to Measure Potential Energy Produced by Solar Panels (IOP Publishing)
- **TOEFL results** : 613/677

4-Year Vocational High School Program in Electronics Engineering – Communication Electronics

Specialization

State Vocational School 1 Cimahi

July 2014 - May 2018

- **Grade Achieved** : 81/100
- **Relevant Coursework** : Electronics, Electrical Systems, Industrial Automation, Communication Systems, and Instrumentation. Electronic Measurement, Equipment Troubleshooting, System Maintenance, and Technical Documentation.

WORK EXPERIENCE AND PROJECTS

Freelance Robotics & Embedded Systems Engineer – Bandung, Indonesia

Self-Employed | September 2023 – Present

- Developed embedded systems and PoC solutions using Arduino, ESP32, and Jetson Nano for automation applications.
- Conducted system integration, testing, troubleshooting, and technical documentation to enhance reliability and performance.

Robotics Research Engineer (Capstone Project) – Bandung, Indonesia

Faculty of Electrical Engineering, Telkom University | January 2024 – August 2024

- Developed and integrated robotic systems involving sensor fusion, embedded controllers, motor control, and PLC-based automation.
- Conducted system validation, troubleshooting, and performance analysis while collaborating with multidisciplinary teams.

Research Intern (Engineering Systems) BRIN – Bandung, Indonesia

Badan Riset dan Inovasi Nasional (BRIN) | June 2023 – August 2023

- Developed and integrated robotic systems using sensor fusion, embedded controllers, and PLC-based automation for prototype development.
- Performed system testing, validation, troubleshooting, and technical documentation to improve system functionality and reliability.

Maintenance Technician – Karawang, Indonesia

PT NT Piston Ring Indonesia | October 2018 - October 2019

- Performed preventive maintenance and troubleshooting of electrical, electronic, and automation systems to improve equipment reliability and minimize downtime.
- Conducted root cause analysis and implemented corrective actions to optimize system performance and operational efficiency.

Mechanical & Electrical Technician– Cimahi, Indonesia

PT Honkys Miniatur Jaya | February 2018 - August 2018

- Performed maintenance, troubleshooting, and repair of electrical and mechanical production systems to ensure operational reliability and efficiency.

On-the-Job Training (Electrical & Mechanical)– Jakarta, Indonesia

PT Sony Indonesia | September 2017 - December 2017

- Assisted in troubleshooting and maintenance of production equipment and electrical systems.
- Supported technical documentation and preventive maintenance activities.

ADDITIONAL EXPERIENCE

Researcher-Assistant Lecturer (Paper/Conference) – Bandung, Indonesia

Engineering and Sciences - International Conference 2024 | January 2024 - September 2024

- Supported technical operations and troubleshooting during conference sessions.
- Assisted engineering discussions and technical presentation activities.

ORGANIZATIONAL EXPERIENCE

Electronics and Intelligence Robotic Research Group, Telkom University

- November 2021 - November 2022 | External Member : Contribute to research and development of robotics technology.
- November 2021 - November 2022 | Member of Abu Indonesia Robot Contest 2022 Telkom University : Design and develop robot competitions.
- November 2022 - November 2023 | External Chair : Lead robotics research projects and manage industrial cooperation.
- November 2022 - November 2023 | Team Leader Abu Indonesia Robot Contest 2023 Telkom University : Lead the robot development team, manage the budget, and design competition strategies.

Community Service Learning, Telkom University

- August 2023 - August 2023 | Volunteer: Participate in community engagement projects, providing support to local initiatives focused on sustainability and development.

TECHNICAL SKILLS & SOFT SKILLS

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| <ul style="list-style-type: none">• Embedded Systems (Arduino, ESP32, Jetson Nano)• Electronics Design, Circuit Analysis & PCB Design• System Integration, Testing & Validation• Sensors, Actuators & IoT Systems• Communication Protocols (UART, I2C, SPI, MQTT, TCP/IP) | <ul style="list-style-type: none">• Instrumentation & Control Systems• Technical Documentation & Reporting <p>Soft Skills : Analytical Thinking & Problem-Solving, Attention to Detail & Troubleshooting, Team Collaboration & Communication, Adaptability & Continuous Improvement Mindset</p> |
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CERTIFICATION AND TRAINING

- Huawei TalentOnline Certifications: Big Data V3.0, Artificial Intelligence (AI) V3.0, Internet of Things (IoT) V2.5, 5G V2.0, Data Center V1.5
- Dicoding Indonesia Certifications: Procedural Programming with Python, Fundamentals of Data Science, Fundamentals of SQL, Fundamentals of Data Visualization, Machine Learning
- Cisco Networking Academy Certifications: Data Science, Cybersecurity

ACHIEVEMENTS AND AWARDS

- 2nd Place of Indonesian Abu Robot Contest Region 1 2022, BPTI, Puspresnas, Indonesia.
- Top 8 National Finalists of Indonesian Abu Robot Contest 2022, BPTI, Puspresnas, Indonesia.
- National Finalists of Indonesian Abu Robot Contest 2023, BPTI, Puspresnas, Indonesia.
- 1st Place of Capstone Design Faculty of Electrical Engineering Telkom University 2024.